PRASANNA REDDY PULAKURTHI

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Education

Rochester Institute of Technology — Rochester, NY
Doctor of Philosophy (Ph.D.) – Electrical and Computer Engineering | ***GPA: 3.93/4.0*** — *Aug 2019 – Aug 2025 (expected)*

Rochester Institute of Technology — Rochester, NY
Master of Science (MS) – Electrical Engineering | ***GPA: 4.0/4.0*** — *Aug 2017 – Jul 2019*

PESIT Bangalore South Campus — Bangalore, India
Bachelor of Engineering (BE) – Electronics and Communications Engineering — *Aug 2013 – Jun 2017*

Research and Work Experience

Qualcomm Internship – Virtual Reality (VR) Team
Engineering Intern — *May 2021 – Aug 2021*

- Developed a novel temperature-compensation algorithm enhancing the accuracy of the 6DOF 4Tx ultrasound VR controller.
- Implemented the algorithm in C, improving the root mean squared error (RMSE) by **11.29%** on synthetic log data.

Microsoft Internship – Iris Recognition Team for HoloLens 2 (Augmented Reality (AR))
Engineering Intern — *May 2020 – Aug 2020*

- Optimized Gabor filter frequencies in Iris Recognition pipeline, significantly enhancing signature uniqueness.
- Developed GAN-based Super Resolution Network to enhance iris image quality, maintaining key Gabor frequencies, improving recognition accuracy on lower-quality images.

Oak Ridge National Laboratory
Research Assistant — *May 2018 – Jul 2019*

- Designed and executed an unmanned aerial system (UAS) data collection to capture aerial imagery of a building casting shadows on 14 differently colored panels, both in and out of shadow. [Link]
- Developed six algorithms for shadow detection in aerial images using color transforms and machine learning techniques, leveraging the collected data. One of the developed chromaticity algorithms improved upon traditional chromaticity approaches by **1.51%** in accuracy and **2.4%** in F-score.

Technical Skills

- **Programming Languages**: C, Python (NumPy, Pandas), HTML/CSS, MATLAB
- **Machine Learning Frameworks/Libraries**: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, LLM, NLP, Transformers
- **Tools**: Docker, DIRSIG, ENVI, Adobe Photoshop, LaTeX, Git, Linux, SQL, Microsoft suite (Excel, Word, PowerPoint)

Publications and Invited Talks

Research Publications

1. **Prasanna Reddy Pulakurthi**, Jiamian Wang, MAJID RABBANI, Sohail Dianat, Raghuveer Rao, Zhiqiang Tao, "X-CoT: Explainable Text-to-Video Retrieval via Large Language Models (LLM) based Chain-of-Thought Reasoning," *EMNLP* 2025.
2. Jiamian Wang, **Prasanna Reddy Pulakurthi**, Pichao WANG, Dongfang Liu, Sohail Dianat, MAJID RABBANI, Raghuveer Rao, Zhiqiang Tao, "Automatic Instruction Programming for Multimodel Large Language Models (MLLM) Based Text-Video Retrieval," *EMNLP* 2025.
3. **Prasanna Reddy Pulakurthi**, Majid Rabbani, Jamison Heard, Sohail Dianat, Celso M. de Melo, and Raghuveer Rao, "Shuffle PatchMix Augmentation with Confidence Margin Weighted Pseudo Labels for Enhanced Source-Free Domain Adaptation," *IEEE International Conference on Image Processing (ICIP)*, 2025. [Link]
4. **Prasanna Reddy Pulakurthi**, Majid Rabbani, Celso M. de Melo, Sohail A. Dianat, and Raghuveer M. Rao. "Effective dual-region augmentation for reduced reliance on large amounts of labeled data," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications III*. SPIE, 2025. [Link]
5. **Prasanna Reddy Pulakurthi**, Mahsa Mozaffari, Sohail Dianat, Jamison Heard, Raghuveer Rao, and Majid Rabbani, "Enhancing GANs with MMD Neural Architecture Search, PMish Activation Function and Adaptive Rank Decomposition," *IEEE Access*, 2024, doi: 10.1109/ACCESS.2024.3485557. [Link]

6. **Prasanna Reddy Pulakurthi**, Celso M De Melo, Raghuveer Rao, and Majid Rabbani, "Enhancing human action recognition with GAN-based data augmentation," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications II.* Vol. 13035. SPIE, 2024. [Link]
7. **Prasanna Reddy Pulakurthi**, Mahsa Mozaffari, Sohail A. Dianat, Majid Rabbani, Jamison Heard, and Raghuveer Rao, "Enhancing GAN Performance through Neural Architecture Search and Tensor Decomposition," *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. [Link]
8. **Prasanna Reddy Pulakurthi**, Sohail A. Dianat, Majid Rabbani, Suya You, and Raghuveer M. Rao, "A globally optimal fast iterative linear maximum likelihood classifier," *Electronic Imaging*, 2023, pp 172-1 - 172-5. [Link]
9. **Prasanna Reddy Pulakurthi**, Sohail A. Dianat, Majid Rabbani, Suya You, and Raghuveer M. Rao, "Unsupervised domain adaptation using feature aligned maximum classifier discrepancy," Applications of Machine Learning. SPIE, 2022. [Link]
10. **Prasanna Reddy Pulakurthi**, "Shadow Detection in Aerial Images using Machine Learning" (2019), Thesis, Rochester Institute of Technology. [Link]

Workshops

1. Sohail Dianat, **Prasanna Reddy Pulakurthi**, Suya You, and Raghuveer M. Rao, "Domain Adaptation using Maximum Mean Discrepancy," poster presented at 2021 Western New York Image and Signal Processing Workshop. [Link]
2. **Prasanna Reddy Pulakurthi**, and Emmett Ientilucci (2019, Feb), "Shadow Detection in Aerial and UAS data Using Neural Networks," presented a talk at STRATUS 2019 Workshop, Rochester, NY, (2019). [Link]

Invited Talks

1. Majid Rabbani and **Prasanna Reddy Pulakurthi**, "Super-Resolution: Computational and Deep Learning-Based Approaches," co-presented a talk at the Society for Imaging Science and Technology (IS&T) Rochester NY chapter. [Link]
2. Raghuveer Rao, **Prasanna Reddy Pulakurthi**, Sohail A. Dianat, invited guest talk at IEEE CIS and C Lecture on "Advancements and Applications of Generative Adversarial Networks." [Link]

PROJECTS

Automatic Attendance Update System using Image Processing [Link]

- Developed a real-time automatic attendance system using OpenCV for face detection, Python-implemented Binarized Statistical Image Features for face recognition, and a Kivy-based app interfacing via sockets and PHP with an Apache-hosted GUI displaying attendance data stored in MySQL.

Automatic License Plate Detection and Recognition [Link]

- Built an automatic number plate detection and recognition system using edge and histogram-based image processing methods for detection and neural networks for recognition, coded in MATLAB.

Machine Learning and AI Projects

- Applied Generative Adversarial Networks (GANs) to unsupervised domain adaptation, image-to-image translation, and human action recognition tasks. [Link]
- Developed Invertible Neural Networks (INNs) to perform super-resolution and image denoising. [Link]
- Designed a full-stack website Vision Captioning Assistant (V-CAT) AI [Link]
- Built a full-stack website for ASL Sign Tutoring (Signmentor AI) [Link]

Image and Signal Processing Projects

- Implemented several image and signal processing algorithms, such as image resizing, noise filtering, image enhancement, image segmentation, camera calibration, stereo reconstruction, affine, and projective transforms.

COURSEWORK

- **Signal & Image Processing**: Digital Signal Processing (EEE678), Digital Image Processing (EEE779), Multi-view Imaging (IMGS712)
- **Machine Learning**: Deep Learning Independent Study (EEE799), AI Explorations (EEE 647), Pattern Recognition (EEE670)
- **Communications**: Information Theory Coding (EEE794), Image and Video Compression (EEE781)
- **Machine Learning Course**: Machine Learning by Stanford University on Coursera